

ARRAYS

Aim:

To understand and implement array operations in Java.

PRE LAB EXERCISE

QUESTIONS

- ✓ What is an array?

An array is a collection of elements of the same data type stored in contiguous memory locations. Each element can be accessed using an index value.

- ✓ Why are arrays used?

Arrays are used to:

Store multiple values of the same type using a single variable

Reduce code complexity

Allow fast access to elements using index

Improve memory organization and efficiency

- ✓ What is the difference between array and variable?

Variable

Stores only one value

Uses a single memory location

Accessed by variable name

Simple data storage

Array

Stores multiple values

Uses multiple contiguous memory locations

Accessed using index

Used for bulk data handling

IN LAB EXERCISE

Objective:

To perform array operations using simple programs.

PROGRAMS:

1. Program to Read and Print Array Elements

Code:

```
import java.util.Scanner;

public class ReadPrintArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[5];
        System.out.println("Enter 5 elements:");
        for(int i = 0; i < 5; i++)
            arr[i] = sc.nextInt();
        System.out.println("Array elements are:");
        for(int i = 0; i < 5; i++)
            System.out.print(arr[i] + " ");
    }
}
```

OUTPUT:

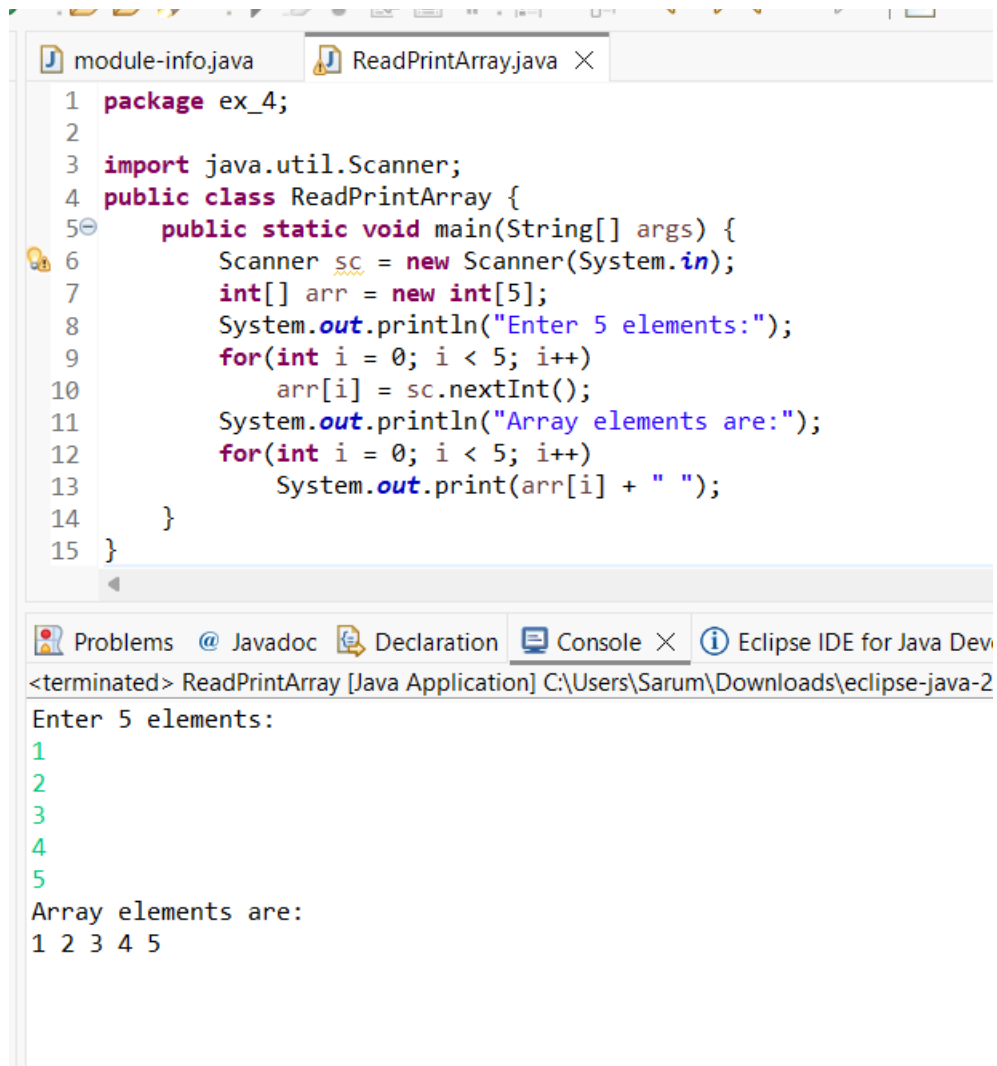
Input:

10 20 30 40 50

Output:

Array elements are:

10 20 30 40 50



The screenshot shows the Eclipse IDE interface. The top editor displays the source code for `ReadPrintArray.java`. The code defines a package `ex_4`, imports `java.util.Scanner`, and defines a public class `ReadPrintArray` with a `main` method. The `main` method uses a `Scanner` to read 5 integers from the user, stores them in an array, and then prints the array elements. The bottom part of the IDE shows the `Console` tab, which displays the program's execution output. The output shows the prompt "Enter 5 elements:" followed by five lines of input (1, 2, 3, 4, 5) and then the prompt "Array elements are:" followed by the output "1 2 3 4 5".

```
1 package ex_4;
2
3 import java.util.Scanner;
4 public class ReadPrintArray {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7         int[] arr = new int[5];
8         System.out.println("Enter 5 elements:");
9         for(int i = 0; i < 5; i++)
10             arr[i] = sc.nextInt();
11         System.out.println("Array elements are:");
12         for(int i = 0; i < 5; i++)
13             System.out.print(arr[i] + " ");
14     }
15 }
```

Problems @ Javadoc Declaration Console X Eclipse IDE for Java Dev

<terminated> ReadPrintArray [Java Application] C:\Users\Sarum\Downloads\eclipse-java-2

Enter 5 elements:

1
2
3
4
5

Array elements are:

1 2 3 4 5

2. Program to Find Sum of Array Elements

Code:

```
import java.util.Scanner;

public class SumArray {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];

        int sum = 0;

        System.out.println("Enter 5 elements:");

        for(int i = 0; i < 5; i++)

            arr[i] = sc.nextInt();

        for(int i = 0; i < 5; i++)

            sum += arr[i];

        System.out.println("Sum = " + sum);

    }

}
```

OUTPUT:

Input:

5 10 15 20 25

Output:

Sum = 75

The screenshot shows the Eclipse IDE interface. The top editor displays the file `SumArray.java` with the following code:

```
1 package ex_4;
2
3 import java.util.Scanner;
4 public class SumArray {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7         int[] arr = new int[5];
8         int sum = 0;
9         System.out.println("Enter 5 elements:");
10        for(int i = 0; i < 5; i++)
11            arr[i] = sc.nextInt();
12        for(int i = 0; i < 5; i++)
13            sum += arr[i];
14        System.out.println("Sum = " + sum);
15    }
16 }
17
```

The bottom of the IDE shows the `Console` tab with the following output:

```
<terminated> SumArray [Java Application] C:\Users\Sarum\Downloads\eclipse-ja
Enter 5 elements:
5
10
15
20
25
Sum = 75
```

3. Program to Find Largest Element in an Array

Code:

```
import java.util.Scanner;

public class LargestElement {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];

        System.out.println("Enter 5 elements:");

        for(int i = 0; i < 5; i++)

            arr[i] = sc.nextInt();

        int max = arr[0];

        for(int i = 1; i < 5; i++)

            if(arr[i] > max)

                max = arr[i];

        System.out.println("Largest element = " + max);

    }

}
```

OUTPUT:

Input:

12 45 23 9 30

Output:

Largest element = 45

The screenshot shows the Eclipse IDE with two tabs: 'module-info.java' and 'LargestElement.java'. The 'LargestElement.java' tab is active, displaying the following Java code:

```
1 package ex_4;
2
3 import java.util.Scanner;
4 public class LargestElement {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7         int[] arr = new int[5];
8         System.out.println("Enter 5 elements:");
9         for(int i = 0; i < 5; i++)
10             arr[i] = sc.nextInt();
11         int max = arr[0];
12         for(int i = 1; i < 5; i++)
13             if(arr[i] > max)
14                 max = arr[i];
15         System.out.println("Largest element = " + max);
16     }
17 }
```

Below the code editor, the 'Console' tab is active, showing the program's output:

```
<terminated> LargestElement [Java Application] C:\Users\Sarum\Downloads\eclipse-
Enter 5 elements:
12
45
23
9
30
Largest element = 45
```

4. Program to Reverse an Array

Code:

```
import java.util.Scanner;

public class ReverseArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[5];
        System.out.println("Enter 5 elements:");
        for(int i = 0; i < 5; i++)
            arr[i] = sc.nextInt();
        System.out.println("Reversed array:");
        for(int i = 4; i >= 0; i--)
            System.out.print(arr[i] + " ");
    }
}
```

OUTPUT:

Input:

1 2 3 4 5

Output:

Reversed array:

5 4 3 2 1


```
1 package ex_4;
2
3 import java.util.Scanner;
4
5 public class ReverseArray {
6     public static void main(String[] args) {
7         Scanner sc = new Scanner(System.in);
8         int[] arr = new int[5];
9         System.out.println("Enter 5 elements:");
10        for(int i = 0; i < 5; i++)
11            arr[i] = sc.nextInt();
12        System.out.println("Reversed array:");
13        for(int i = 4; i >= 0; i--)
14            System.out.print(arr[i] + " ");
15    }
16 }
17
18
```

Problems @ Javadoc Declaration Console X Eclipse ID

<terminated> ReverseArray [Java Application] C:\Users\Sarum\Downloads\ex

Enter 5 elements:

1
2
3
4
5

Reversed array:
5 4 3 2 1

5. Program to Count Even and Odd Numbers

Code:

```
import java.util.Scanner;

public class EvenOddCount {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];

        int even = 0, odd = 0;

        System.out.println("Enter 5 elements:");

        for(int i = 0; i < 5; i++)

            arr[i] = sc.nextInt();

        for(int i = 0; i < 5; i++) {

            if(arr[i] % 2 == 0)

                even++;

            else

                odd++;

        }

        System.out.println("Even = " + even);

        System.out.println("Odd = " + odd);

    }

}
```

OUTPUT:

Input:

2 7 4 9 10

Output:

Even = 3

Odd = 2

The screenshot shows the Eclipse IDE interface. The top editor displays the source code for `EvenOddCount.java`. The code is as follows:

```
1 package ex_4;
2
3 import java.util.Scanner;
4 public class EvenOddCount {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7         int[] arr = new int[5];
8         int even = 0, odd = 0;
9         System.out.println("Enter 5 elements:");
10        for(int i = 0; i < 5; i++)
11            arr[i] = sc.nextInt();
12        for(int i = 0; i < 5; i++) {
13            if(arr[i] % 2 == 0)
14                even++;
15            else
16                odd++;
17        }
18    }
19 }
```

The bottom editor shows the console output:

```
<terminated> EvenOddCount [Java Application] C:\Users\Sarum\Downloads\eclipse-ja
Enter 5 elements:
1
2
3
4
5
Even = 2
Odd = 3
```

6. Program to Sort Array in Ascending Order

Code:

```
import java.util.Scanner;

public class SortArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];
        int temp;

        System.out.println("Enter 5 elements:");
        for(int i = 0; i < 5; i++)
            arr[i] = sc.nextInt();
        for(int i = 0; i < 5; i++) {
            for(int j = i + 1; j < 5; j++) {
                if(arr[i] > arr[j]) {
                    temp = arr[i];
                    arr[i] = arr[j];
                    arr[j] = temp;
                }
            }
        }
        System.out.println("Sorted array:");
        for(int i = 0; i < 5; i++)
            System.out.print(arr[i] + " ");
    }
}
```

OUTPUT:

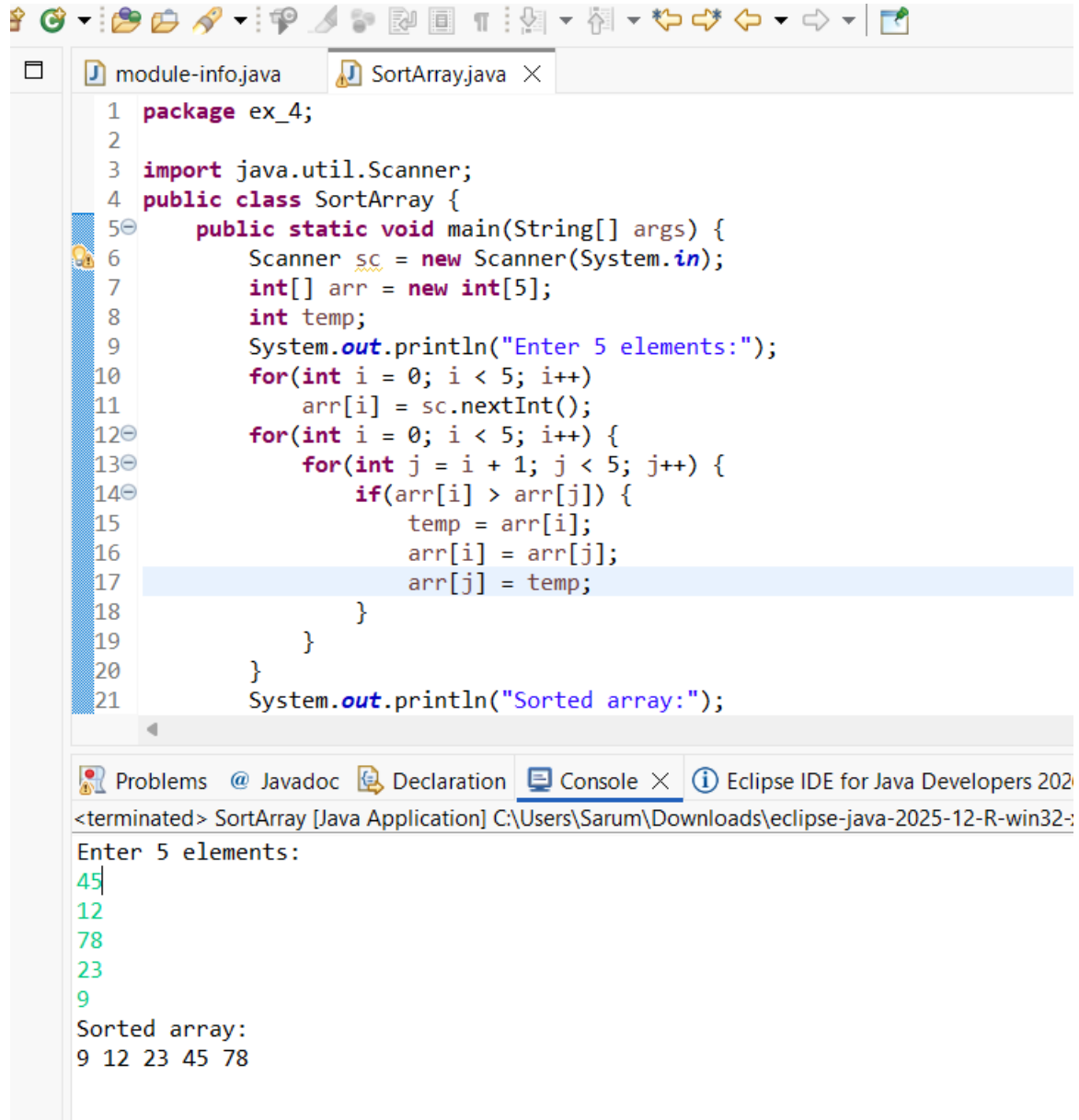
Input:

45 12 78 23 9

Output:

Sorted array:

9 12 23 45 78



The screenshot shows the Eclipse IDE interface. The top toolbar contains various icons for file operations and development tools. Below the toolbar, the 'Package Explorer' on the left shows a project named 'ex_4' containing two files: 'module-info.java' and 'SortArray.java'. The 'SortArray.java' file is open in the editor, displaying the following Java code:

```
1 package ex_4;
2
3 import java.util.Scanner;
4 public class SortArray {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7         int[] arr = new int[5];
8         int temp;
9         System.out.println("Enter 5 elements:");
10        for(int i = 0; i < 5; i++)
11            arr[i] = sc.nextInt();
12        for(int i = 0; i < 5; i++) {
13            for(int j = i + 1; j < 5; j++) {
14                if(arr[i] > arr[j]) {
15                    temp = arr[i];
16                    arr[i] = arr[j];
17                    arr[j] = temp;
18                }
19            }
20        }
21        System.out.println("Sorted array:");
```

Below the editor, the 'Console' view is open, showing the output of the program:

```
<terminated> SortArray [Java Application] C:\Users\Sarum\Downloads\eclipse-java-2025-12-R-win32-;
Enter 5 elements:
45
12
78
23
9
Sorted array:
9 12 23 45 78
```

7. Program to Find Second Largest Element

Code:

```
import java.util.Scanner;

public class SecondLargest {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];

        System.out.println("Enter 5 elements:");

        for(int i = 0; i < 5; i++)

            arr[i] = sc.nextInt();

        int largest = arr[0];

        int second = arr[0];

        for(int i = 0; i < 5; i++) {

            if(arr[i] > largest) {

                second = largest;

                largest = arr[i];

            }

        }

        System.out.println("Second largest = " + second);

    }

}
```

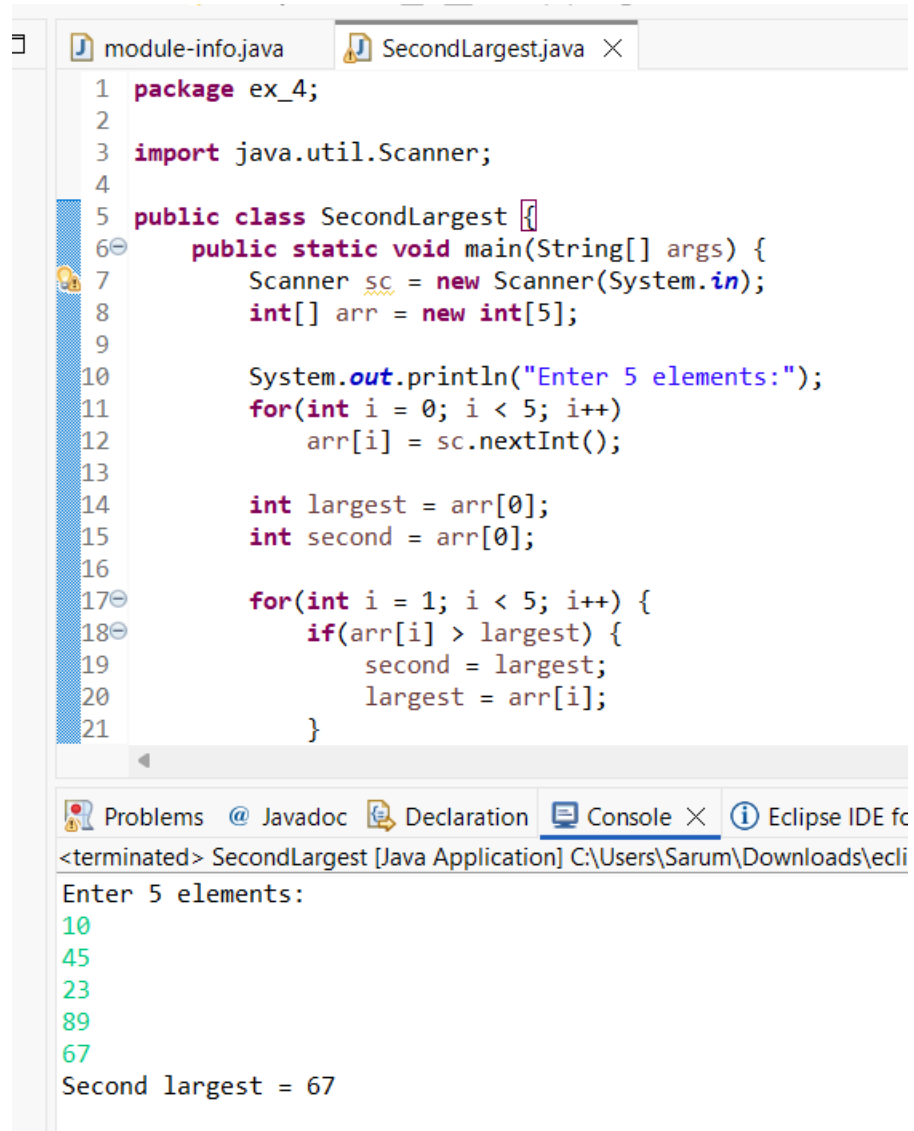
OUTPUT:

Input:

10 45 23 89 67

Output:

Second largest = 67



The screenshot shows the Eclipse IDE with two tabs: 'module-info.java' and 'SecondLargest.java'. The 'SecondLargest.java' tab is active, displaying the following code:

```
1 package ex_4;
2
3 import java.util.Scanner;
4
5 public class SecondLargest {
6     public static void main(String[] args) {
7         Scanner sc = new Scanner(System.in);
8         int[] arr = new int[5];
9
10        System.out.println("Enter 5 elements:");
11        for(int i = 0; i < 5; i++)
12            arr[i] = sc.nextInt();
13
14        int largest = arr[0];
15        int second = arr[0];
16
17        for(int i = 1; i < 5; i++) {
18            if(arr[i] > largest) {
19                second = largest;
20                largest = arr[i];
21            }
22        }
23    }
24 }
```

Below the code editor, the 'Console' tab is active, showing the program's execution output:

```
<terminated> SecondLargest [Java Application] C:\Users\Sarum\Downloads\ecli
Enter 5 elements:
10
45
23
89
67
Second largest = 67
```

8. Program for Matrix Addition (2D Array)

Code:

```
import java.util.Scanner;

public class MatrixAddition {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[][] a = new int[2][2];
        int[][] b = new int[2][2];
        int[][] sum = new int[2][2];

        System.out.println("Enter elements of matrix A:");

        for(int i = 0; i < 2; i++)
            for(int j = 0; j < 2; j++)
                a[i][j] = sc.nextInt();

        System.out.println("Enter elements of matrix B:");

        for(int i = 0; i < 2; i++)
            for(int j = 0; j < 2; j++)
                b[i][j] = sc.nextInt();

        for(int i = 0; i < 2; i++)
            for(int j = 0; j < 2; j++)
                sum[i][j] = a[i][j] + b[i][j];

        System.out.println("Sum matrix:");

        for(int i = 0; i < 2; i++) {
            for(int j = 0; j < 2; j++)
                System.out.print(sum[i][j] + " ");

            System.out.println();
        }
    }
}
```



```
}  
}  
}
```

OUTPUT:

Matrix A:

1 2

3 4

Matrix B:

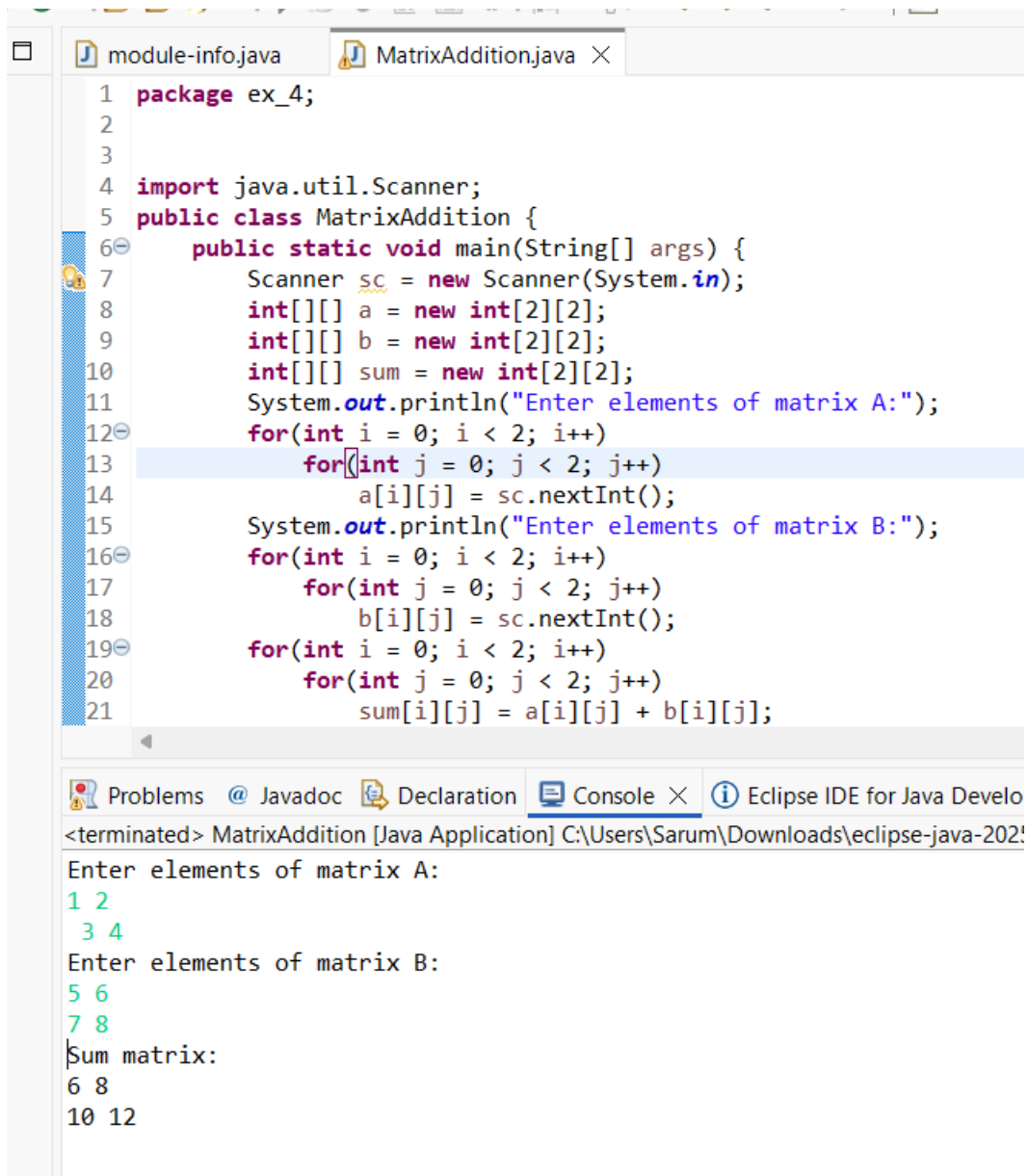
5 6

7 8

Sum matrix:

6 8

10 12



```
1 package ex_4;
2
3
4 import java.util.Scanner;
5 public class MatrixAddition {
6     public static void main(String[] args) {
7         Scanner sc = new Scanner(System.in);
8         int[][] a = new int[2][2];
9         int[][] b = new int[2][2];
10        int[][] sum = new int[2][2];
11        System.out.println("Enter elements of matrix A:");
12        for(int i = 0; i < 2; i++)
13            for(int j = 0; j < 2; j++)
14                a[i][j] = sc.nextInt();
15        System.out.println("Enter elements of matrix B:");
16        for(int i = 0; i < 2; i++)
17            for(int j = 0; j < 2; j++)
18                b[i][j] = sc.nextInt();
19        for(int i = 0; i < 2; i++)
20            for(int j = 0; j < 2; j++)
21                sum[i][j] = a[i][j] + b[i][j];
```

Problems @ Javadoc Declaration Console Eclipse IDE for Java Develo

<terminated> MatrixAddition [Java Application] C:\Users\Sarum\Downloads\eclipse-java-202!

Enter elements of matrix A:

1 2
3 4

Enter elements of matrix B:

5 6
7 8

Sum matrix:

6 8
10 12

POST LAB EXERCISE

- ✓ Why is array indexing usually started from zero instead of one?
Array indexing starts from zero because the index represents the offset from the base memory address. The first element is stored at the base address itself, so its offset is zero.
- ✓ What happens if we try to access an array element outside its declared size?
If we access an array element outside its size, Java throws an `ArrayIndexOutOfBoundsException` at runtime.
- ✓ How does memory allocation differ for static arrays and dynamic arrays?
Static arrays: Size is fixed at compile time and cannot be changed.
Dynamic arrays: Size is decided at runtime and can be changed (e.g., using `ArrayList`).
- ✓ Why is searching faster in arrays compared to linked lists?
Arrays support direct access using index, so elements can be accessed in $O(1)$ time.
Linked lists require sequential traversal, which takes more time.
- ✓ What is the difference between contiguous and non-contiguous memory allocation?

Contiguous Memory

Memory blocks are adjacent

Used in arrays

Faster access

Fixed size

Non-Contiguous Memory

Memory blocks are scattered

Used in linked lists

Slower access

Dynamic size

Result:

Thus the array operations were executed successfully.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		